Chapter 1: Vector Spaces

Linear Algebra Done Right, by Sheldon Axler

A: $\mathbb{R}$ and $\mathbb{C}$

Problem 1

Verify that the definition of equality on the integers is both reflexive and symmetric

Proof. First, we prove that the equality is reflexive. Suppose $a \text{---} b = a' \text{---} b'$. Then, $a + b' = a' + b$. Because the equality on the natural numbers is reflexive, we have that $a' + b = a + b'$. By definition of the equality on the integers, we have that $a' \text{---} b' = a \text{---} b$.

Next, we show that the equality on the integers is symmetric. Suppose $a \text{---} b = c \text{---} d$ and $c \text{---} d = e\, minus f$. Then, $a + c = d + b$ and $e + d = c + f$. We add the second equation into the first to get:

$$a + c + e + d = d + b + c + f.$$

By the cancellation law of natural numbers, we have:

$$a + e = b + f.$$

By the definition of equality on the integers, $a \text{---} b = f \text{---} e$. ☐

Problem 2

Find two distinct square roots of i.

Proof. Suppose $a, b \in \mathbb{R}$ are such that $(a + bi)^2 = i$. Then

$$(a^2 - b^2) + (2ab)i = i.$$

Since the real and imaginary part of both sides must be equal, respectively, we have a system of two equations in two variables

$$\begin{aligned} a^2 - b^2 &= 0 \\ ab &= \frac{1}{2}. \end{aligned}$$

The first equation implies $b = \pm a$. Plugging $b = -a$ into the second equation would imply $-a^2 = 1/2$, which is impossible, and hence we must have $a = b$. But this in turn tells us $a = \pm 1/\sqrt{2}$, and hence our two roots are

$$\pm \left(\frac{1}{\sqrt{2}} \right) (1 + i),$$

as desired. ☐

B: Definition of a Vector Space

C: Subspaces